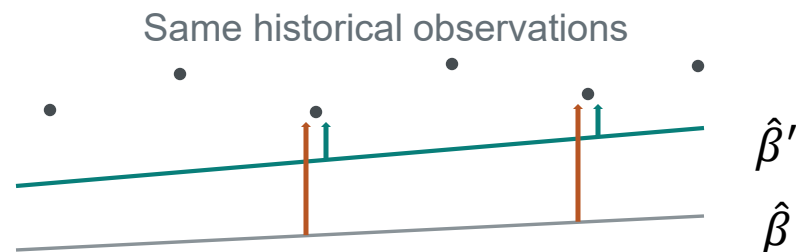


(a) Revision-aware memory

What must survive compression?



Residual-only memory becomes stale

$$\Delta h_u = - \boxed{R_{u\beta}} \Delta \beta$$

required cross information

Residual-only memory

✗ stale

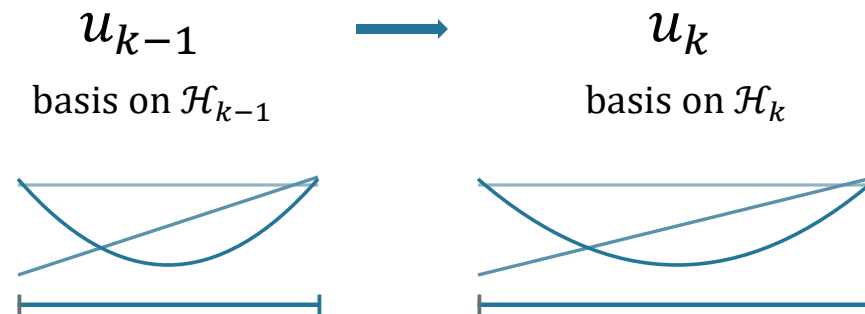
Joint likelihood memory

✓ revision-aware

**Cross information enables
replay-free revision.**

(b) Coordinate-consistent transfer

How does historical evidence move?



Same dimension, different GP functionals

Conditional projection

$$\mathbb{E}[u_{k-1} | u_k] = L_{k-1,k} u_k$$

Likelihood transport

$$R_k^- = M^\top R_{k-1} M$$

$$r_k^- = M^\top r_{k-1}$$

**Transport likelihood evidence,
not latent states.**

L^2 -optimal finite projection;
not generally lossless.

(c) Structure-preserving recovery

Why does inference remain scalable?

Single-Kronecker invariant

$$R_{uu,k} = B_k \otimes G$$



$$D_u = K_{t,k}^{-1} \otimes K_s^{-1} + B_k \otimes G$$

prior + transported likelihood



Schur + Sylvester



$$q_k(\beta, u_k)$$

Exact recovery of the transferred finite target
without dense product-state factorisation

$$M_s \simeq M_t = M:$$

$O(M^3)$ recovery; $O(M^2)$ persistent state